Safe RAN Slicing in O-RAN: Minimizing SLA Violations via Model-Based Reinforcement Learning

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***Abstract*—RAN slicing in Open RAN (O-RAN) architectures requires dynamic and intelligent resource allocation to ensure both high resource efficiency and strict compliance with Service Level Agreements (SLAs). However, conventional reinforcement learning (RL) methods struggle in this setting due to their unsafe exploration behavior, often resulting in frequent SLA violations. In this paper, we propose a safe and sample-efficient model-based RL (MBRL) framework tailored for SLA-aware RAN slicing. Our agent leverages an online kernelized quantile regressor to estimate the lower quantile of key performance indicators (KPIs), enabling direct modeling of SLA violation risk. To ensure safety during online adaptation, the policy incorporates an uncertainty-aware safety margin, promoting conservative decisions under high model uncertainty. Extensive simulations demonstrate that our method reduces SLA violations to under 1%, while maintaining high resource efficiency and achieving fast, stable convergence.**

***Index Terms*—O-RAN, RAN Slicing, Safe Resource Allocation, Online Learning, Quantile Regression, Kernel Method.**

I. Introduction

The advent of 5G and the evolution towards 6G networks promise to support a diverse ecosystem of services far beyond simple mobile broadband [1]. These services, including enhanced Mobile Broadband (eMBB) for high-definition streaming, Ultra-Reliable Low-Latency Communications (URLLC) for autonomous systems and the tactile internet, and massive Machine Type Communications (mMTC) for the Internet of Things, present conflicting performance requirements. A single, monolithic network architecture cannot efficiently serve these disparate needs simultaneously. Network slicing has emerged as the key architectural solution, enabling network operators to create multiple virtual, logically-isolated end-to-end networks on top of a common physical infrastructure. Each slice can be independently configured and optimized for the specific demands of the service it supports [2].

To realize the full potential of network slicing, the underlying network must be open, programmable, and intelligent. The Open Radio Access Network (O-RAN) paradigm addresses this need by disaggregating the RAN and introducing a new logical node, the RAN Intelligent Controller (RIC) [3]. The RIC is designed to host third-party applications (known as xApps and rApps) that can leverage AI and machine learning to manage and optimize network resources in near-real-time, enabling unprecedented levels of automation and efficiency [4]. This shift from static configuration to intelligent, data-driven control is fundamental to delivering the flexibility promised by network slicing.

With this new architectural capability comes a critical challenge: one of the most vital functions for an xApp in the RIC is the dynamic allocation of radio resources, such as Physical Resource Blocks (PRBs) [5], among the various network slices. This task embodies a difficult trade-off. On one hand, the controller must guarantee that each slice receives sufficient resources to meet its unique and strict Service Level Agreement (SLA). On the other hand, for commercial viability, it must maximize overall spectral efficiency by not over-provisioning resources. This becomes the central problem: how to design an intelligent agent that can learn an optimal resource allocation policy online that satisfies both of these conflicting objectives.

Reinforcement Learning (RL) has emerged as a powerful tool for dynamic decision-making in complex network environments. However, traditional model-free RL (MFRL) methods—such as Q-learning and PPO—rely inherently on extensive trial-and-error exploration, which not only leads to unsafe policy updates but also results in frequent and severe violations of SLAs. Moreover, the high sample complexity and computational cost of MFRL hinder its applicability in resource-constrained RIC platforms. In response, recent works have turned to MBRL for improved sample efficiency and safer online adaptation. Yet, much of the existing MBRL literature still relies on overly simplistic models or offline assumptions, often lacking mechanisms to quantify uncertainty or explicitly minimize SLA risks.

To this end, we propose a novel model-based reinforcement learning (MBRL) agent with online quantile regression designed specifically for safe and sample-efficient operation within the RIC. Our approach learns a probabilistic model of the system's performance to directly estimate the risk of an SLA violation for any given resource allocation. The agent's control policy then leverages this learned model to make risk-aware decisions, acting with caution when faced with uncertainty to ensure safe exploration and stable performance. This allows the system to directly target a desired SLA violation

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rate while concurrently optimizing for resource efficiency.

The primary contributions of this work are as follows:

- We propose a novel MBRL agent architecture suitable for deployment in an O-RAN RIC, designed for safe online learning in RAN slicing.
- We introduce the use of online quantile regression as the core learning mechanism, allowing the agent to directly model and manage the risk of SLA violations.
- We design a risk-aware control policy that leverages model uncertainty to apply a dynamic security margin, creating a system that balances safe exploration with resource efficiency.

The rest of the paper is organized as follows. Section II discusses the related work. Section III and Section IV formulate the problem and proposed solution. Section V evaluates the proposal with system simulation and compares with related work. Section VI briefly concludes this paper.

II. Related Work

The dynamic nature of resource management in 5G and beyond has driven a surge of interest in applying RL. In the context of RAN slicing, most existing work adopts MFRL approaches due to their strong performance in high-dimensional control tasks.

A dominant trend is the use of Deep Q-Networks (DQN) and their variants for slice resource management. For example, [6] employs a Double Dueling Q-learning agent to control resource sharing and scheduling across services, while [7] introduces a GAN-based Distributional Q-Network (GAN-DDQN) to improve spectrum efficiency and QoE. These methods typically enforce SLA constraints via reward shaping, treating violations as penalties. However, such "soft" enforcement does not prevent the agent from making high-risk decisions that can result in severe SLA violations—an unacceptable outcome for mission-critical services.

Other works explore enhancements in agent design or problem formulation. [8] proposes a twin-timescale hierarchical agent to decouple short- and long-term decisions, while [9] uses an Anytime-Competitive MDP to ensure perpetual SLA guarantees. Methods like MicroOpt [10] and Explanation-Guided DRL [11] shift away from end-to-end RL, employing gradient-based optimization or XAI pipelines for improved transparency. While innovative, these approaches often introduce significant architectural complexity and still rely on reward-penalty mechanisms that lack hard safety guarantees.

Our work takes a different direction by proposing a lightweight, online model-based agent that is both sample-efficient and inherently safe. The most comparable prior method is KBRL by Alcaraz et al. [12], which uses a model-based kernel method to boost sample efficiency. However, KBRL's reliance on a monotonicity assumption limits its applicability in real RAN environments, where physical-layer effects often introduce non-monotonic behaviors. Our agent removes this assumption by learning a more nuanced, risk-aware model of KPI response. This enables safe and efficient learning even under complex and unpredictable conditions, making our approach more robust and broadly applicable than previous model-based methods.

III. Problem Formulation

We address the dynamic allocation of radio resources in a RAN that serves multiple, heterogeneous network slices. The core challenge is to allocate resources efficiently while providing robust probabilistic guarantees for the SLA of each slice in a non-stationary environment.

A. System Model

We consider a system, as shown in the left part of Figure 1, composed of a single base station allocating a total of C PRBs among a set of L coexisting network slices, indexed by $i \in \{1, \dots, L\}$. The system operates in discrete time steps $t = 1, 2, \dots$, where each step represents a decision period over which a resource allocation is held constant. To align with O-RAN specification, we set that each step is 100 TTIs and each TTI is 1 ms. The environment is non-stationary, characterized by network users arriving and departing according to a stochastic process.

B. State and Action Spaces

At the beginning of each time step t, the control agent observes a local state vector $\mathbf{s}_t^{(i)} \in \mathbb{R}^{d_i}$ for each slice i. This vector contains aggregated Key Performance Indicators (KPIs) such as the lowest user throughput, user count, queue lengths, and Signal-to-Noise Ratio (SNR). To enable cooperative behavior, each agent's state is enriched with a mean-field representation of the global system state including global statistics (e.g., total PRB usage of other slices, total number of UEs), $\bar{\mathbf{s}}_t \in \mathbb{R}^{d_g}$, allowing the agent to proactively learn the correlation between system-wide congestion and the need for conservative proposals. The full state representation for agent i is the concatenation $\mathbf{z}_t^{(i)} = [\mathbf{s}_t^{(i)}, \bar{\mathbf{s}}_t]$.

Based on this observation, the agent selects a joint action $\mathbf{a}_t = (a_t^{(1)}, \dots, a_t^{(L)})$, where $a_t^{(i)} \in \{0, 1, \dots, C\}$ is the number of PRBs allocated to slice i. The action must satisfy the capacity constraint: $\sum_{i=1}^{L} a_t^{(i)} \leq C$.

After applying action $\mathbf{a}_t$, the system transitions, and a true performance metric $y_t^{(i)}$ (e.g., the 5th percentile of user throughput) is observed for each slice.

C. Objective: A Constrained Markov Decision Process

The goal is to find a distributed control policy, $\boldsymbol{\pi} = (\pi^{(1)}, \dots, \pi^{(L)})$, that minimizes the total allocated radio resources while ensuring the SLA for each slice is met with high probability. This can be formulated as a Constrained Markov Decision Process (CMDP):

$$\begin{aligned} &\min_{\boldsymbol{\pi}} \lim_{T\to\infty} \frac{1}{T} \mathbb{E}_{\boldsymbol{\pi}} \left[\sum_{t=1}^{T} \sum_{i=1}^{L} a_t^{(i)} \right] \\ &\text{s.t.} \lim_{T\to\infty} \frac{1}{T} \mathbb{E}_{\boldsymbol{\pi}} \left[\sum_{t=1}^{T} \mathbb{I}(y_t^{(i)} < \Lambda^{(i)}) \right] \leq \delta^{(i)};\ \forall i \in L \end{aligned} \tag{1}$$

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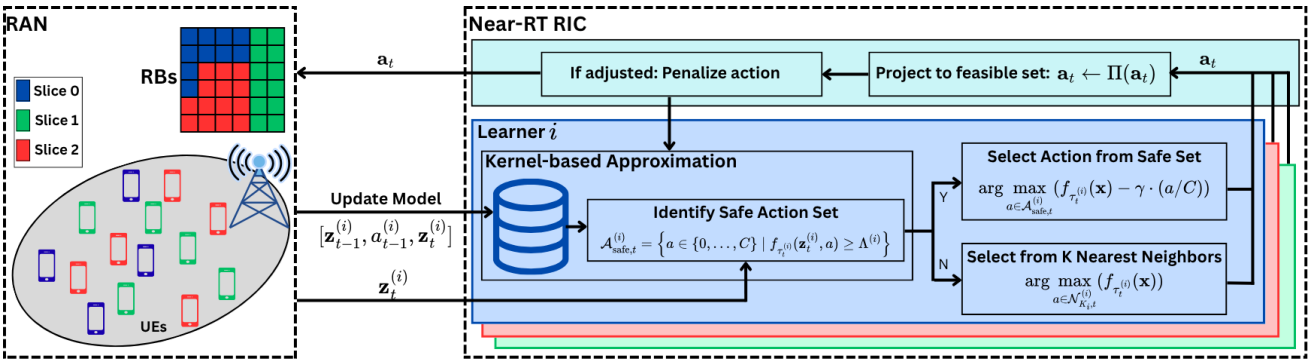


Fig. 1: System

where:

- $\mathbb{E}_{\boldsymbol{\pi}}$ denotes the expectation over all possible system trajectories under the joint policy $\boldsymbol{\pi}$.
- $\Lambda^{(i)}$ is the minimum acceptable KPI value (the SLA threshold) for slice i.
- $\mathbb{I}(\cdot)$ is the indicator function, which is 1 if the condition is true (an SLA violation occurs) and 0 otherwise.
- $\delta^{(i)}$ is the maximum tolerable long-term violation rate for slice i (e.g., 1%).

The transition dynamics of the system are unknown, the state space is large and continuous, and the environment is non-stationary, making this CMDP intractable to solve with traditional methods.

IV. Safe Online Quantile Regression

We propose a distributed, risk-averse control system composed of L independent learning agents. The core of our solution is a novel learning framework that integrates Online Quantile Regression with mechanisms for explicit safety, multi-agent cooperation, and adaptive risk management.

A. Online Quantile Regression Model

For each slice i, we learn a function $f_{\tau_t^{(i)}}$ that, given an enriched state $\mathbf{z}_t^{(i)}$ and an action $a_t^{(i)}$, predicts the $\tau_t^{(i)}$-th quantile of the performance KPI $y_t^{(i)}$. The input to the model is the feature vector $\mathbf{x}_t^{(i)} = [\mathbf{z}_t^{(i)}, a_t^{(i)}/C]$.

The model $f_{\tau_t^{(i)}}$ is a kernelized function approximator, defined by a set of B support vectors (landmarks) $\mathbf{x}j$ and their corresponding coefficients α_j, maintained in a memory buffer with a fixed budget:

$$f\tau_t^{(i)}(\mathbf{x}) = \sum_{j=1}^{B} \alpha_j k(\mathbf{x}, \mathbf{x}_j),$$

$$\sigma_t^{(i)}(\mathbf{x}) = 1 - \max_j k(\mathbf{x}, \mathbf{x}_j),$$

where $k(\cdot,\cdot)$ is the Matérn kernel, chosen for its flexibility in modeling complex, nonlinear functions. The term $\sigma_t^{(i)}(\mathbf{x})$ quantifies the uncertainty in the model's approximation.

The agent improves upon the baseline by incorporating an asymmetric, cost-sensitive learning rule [13]. The update for a new support vector's coefficient, $\alpha_{\text{new}}^{(i)}$, is defined as:

$$\begin{cases} \alpha_{\text{UP}}^{(i)} = \eta \cdot \tau_t^{(i)} & \text{if } y_t^{(i)} > f_{\tau_t^{(i)}}(\mathbf{x}) \text{ (Under-prediction)} \\ \alpha_{\text{OP}}^{(i)} = \eta \cdot (\tau_t^{(i)} - 1) & \text{if } y_t^{(i)} \leq f_{\tau_t^{(i)}}(\mathbf{x}) \text{ (Over-prediction)} \end{cases}$$

where η is the learning rate and $\tau_t^{(i)}$ is the target quantile.

In cases of critical failure—where a predicted-safe action leads to an SLA violation—we further define $\alpha_{\text{CF}}^{(i)} = \alpha_{\text{OP}}^{(i)} \cdot P_{\text{penalty}}$, where $P_{\text{penalty}} > 1$ is a penalty amplification factor used to emphasize the severity of such prediction errors.

The model is updated online at each step. When a new observation $(\mathbf{x}_t^{(i)}, y_t^{(i)})$ is received, a new support vector is added with its coefficient computed from the gradient of the pinball loss [13]. To maintain the memory budget B, the least informative support vectors are discarded, prioritizing those that contribute most to learning. Over time, the model converges to a decision boundary that is systematically more risk-averse than $\tau_t^{(i)}$-quantile, as proven in Section IV-D.

B. Hierarchical Control Policy

The policy $\pi^{(i)}$ for each agent selects an action through a multi-stage process designed to prioritize safety while minimizing resource consumption.

1) Safe Set Identification: The agent first identifies the set of actions predicted to satisfy the SLA constraint with confidence level $(1 - \tau_t^{(i)})$. The safe action set is defined as:

$$\mathcal{A}_{\text{safe},t}^{(i)} = \left\{ a \in \{0,\dots,C\} \mid f_{\tau_t^{(i)}}(\mathbf{z}_t^{(i)}, a) \geq \Lambda^{(i)} \right\}$$

2) Action Selection Logic: If $\mathcal{A}_{\text{safe},t}^{(i)}$ is non-empty (Confident Mode), the agent seeks to minimize resource usage while maintaining SLA satisfaction. It selects the action with the highest efficiency-adjusted score:

$$a_t^{(i)} = \arg\max_{a \in \mathcal{A}_{\text{safe},t}^{(i)}} (f_{\tau_t^{(i)}}(\mathbf{z}_t^{(i)}, a) - \gamma \cdot (a/C))$$

where γ is a cost penalty promoting resource efficiency.

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If $\mathcal{A}^{(i)}_{\text{safe},t}$ is empty (Uncertain Mode), the agent falls back to a hierarchical exploratory strategy: With probability ϵ, it samples a Softmax-based exploratory action:

$$a_t^{(i)} \sim P(a) \propto \exp\left(\frac{f_{\tau_t^{(i)}}(\mathbf{z}_t^{(i)}, a)}{T_{\text{softmax}}}\right),$$

where T_{softmax} is the temperature controlling exploration sharpness. With probability $1-\epsilon$, it invokes a Safety-First k-NN heuristic, searching for the k nearest past states and selecting a known successful action (i.e., $y \geq \Lambda^{(i)}$):

$$a_t^{(i)} = \arg\max_{a \in \mathcal{N}^{(i)}_{K_i,t}} (f_{\tau_t^{(i)}}(\mathbf{z}_t^{(i)}, a)),$$

where $\mathcal{N}^{(i)}_{K_i,t}$ is the set of actions executed in the k nearest-neighbor states.

3) Resource Adjustment: To enhance reliability under uncertainty, the agent adjusts each action by a safety margin:

$$a_t^{(i)} \leftarrow a_t^{(i)} + \text{round}(\sigma_t^{(i)} \cdot M_t^{(i)}),$$

where $\sigma_t^{(i)}$ is the model's uncertainty and $M_t^{(i)}$ controls the margin scale. Larger margins are applied in high-uncertainty cases to prevent SLA violations.

If the sum of requested resources exceeds system capacity ($\sum_i a_t^{(i)} > C$), a centralized projection operator Π proportionally scales down the allocation vector to enforce feasibility. If a slice's action is reduced due to this projection, a virtual update is triggered: a support vector representing the original (pre-projection) action is added to the learner's memory with a small negative weight. This penalizes overly aggressive proposals and guides the agent to avoid decisions that frequently require correction.

Algorithm 1 Safe Online Quantile Regression

Input: ϵ, γ, C, K, η.
Initialize: $\{f_{\tau_1^{(i)}}, \tau_1^{(i)}, M_1^{(i)}, \sigma_1^{(i)}\}_{i=1}^{L}$, $B = \emptyset$.
for $t = 1, \ldots, T$ **do**
  for $i = 1, \ldots, L$ **do**
    $\mathcal{A}^{(i)}_{\text{safe},t} = \left\{a \in \{0, \ldots, C\} \mid f_{\tau_t^{(i)}}(\mathbf{z}_t^{(i)}, a) \geq \Lambda^{(i)}\right\}$
    if $\mathcal{A}^{(i)}_{\text{safe},t} \neq \emptyset$ **then**
      $U(a) \leftarrow f_{\tau_t^{(i)}}(\mathbf{z}_t^{(i)}, a) - \gamma \frac{a}{C}$
      $a^{(i)}_{\text{base},t} \leftarrow \arg\max_{a \in \mathcal{A}^{(i)}_{\text{safe},t}} U(a)$
    else
      if random() $< \epsilon$ **then**
        $a^{(i)}_{\text{base},t} \sim \exp(f_{\tau_t^{(i)}}(\mathbf{z}_t^{(i)}, a)/T_{\text{softmax}})$
      else
        $a^{(i)}_{\text{base},t} \leftarrow \arg\max_{a \in \mathcal{N}^{(i)}_{K_i,t}} (f_{\tau_t^{(i)}}(\mathbf{z}_t^{(i)}, a))$
      end if
    end if
    $a^{(i)}_{\text{final},t} \leftarrow a^{(i)}_{\text{base},t} + \text{round}(\sigma_t^{(i)} \cdot M_t^{(i)})$
  end for
  if $\sum_{i=1}^{L} a^{(i)}_{\text{final},t} > C$ **then**
    $\mathbf{a}_{\text{final},t} \leftarrow \Pi(\mathbf{a}_{\text{final},t})$ ▷ Project to feasible set
  end if
  Execute $\mathbf{a}_{\text{final},t}$, observe outcome y_t and next state $\mathbf{s}_t$.
  for $i = 1, \ldots, L$ **do**
    Update $f_{\tau_{t+1}^{(i)}}$ with $(\mathbf{z}_t^{(i)}, a^{(i)}_{\text{final},t}, y_t^{(i)})$ to get $f_{\tau_{t+1}^{(i)}}$.
    Apply adjustment penalty update.
    Update $\tau_{t+1}^{(i)}, M_{t+1}^{(i)}$.
  end for
end for

C. Dynamic Quantile and Margin Adaptation

To adapt its conservatism, the agent dynamically tunes the quantile $\tau_t^{(i)}$ and margin scale $M_t^{(i)}$ per slice via a heuristic controller that tracks recent SLA violations. When the violation rate increases, the controller lowers $\tau_t^{(i)}$ to encourage more pessimistic predictions and increases $M_t^{(i)}$ to add a larger safety margin in terms of RBs. Conversely, when SLA performance is near-perfect, it cautiously raises $\tau_t^{(i)}$ and reduces $M_t^{(i)}$, thereby improving resource efficiency without compromising reliability.

D. Convergence Analysis of the Risk-Averse Learning Rule

Convergence to a fixed numeric value is intractable due to environmental non-stationarity and adaptive agent behavior. However, we prove that the agent consistently converges to a decision boundary that is more risk-averse than the baseline τ-quantile. Our agent builds upon online quantile regression, which updates predictions using the gradient of the pinball loss. The standard formulation guarantees convergence to the true τ-quantile of the conditional distribution $\mathbb{P}(y|x)$ [13], such that under-prediction occurs with probability τ and over-prediction with $1-\tau$. Convergence is achieved when the expected update becomes zero: $E[\alpha_{\text{new}}] = 0$. With our cost-sensitive update rule, the equilibrium condition becomes:

$$\mathbb{P}(y > f(x)) \cdot \alpha_{\text{UP}} + \mathbb{P}(\Lambda \leq y \leq f(x)) \cdot \alpha_{\text{OP}} + \mathbb{P}(y < \Lambda) \cdot \alpha_{\text{CF}} = 0$$

We substitute the update values and rearrange this equation to show the balance on the prediction:

$$\underbrace{\mathbb{P}(y > f(x)) \cdot \tau}_{\text{Total Upward Push}} = \underbrace{\mathbb{P}(\Lambda \leq y \leq f(x)) \cdot (1-\tau)}_{\text{Normal Downward Push}} + \underbrace{\mathbb{P}(y < \Lambda) \cdot (1-\tau) \cdot P_{\text{penalty}}}_{\text{Amplified Penalty Downward Push}}$$

To maintain equilibrium under this stronger downward force, the model must increase the probability of under-prediction, $\mathbb{P}(y > f(x))$. The only way to achieve this is to lower the prediction $f(x)$, thereby shifting the decision boundary in a more conservative direction.

This establishes that the cost-sensitive learning rule induces convergence to a more pessimistic (i.e., safer) decision boundary than the nominal τ-quantile. While the exact final SLA satisfaction rate depends on the environment's stochastic dynamics, the learning process is provably biased toward higher

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TABLE I: Simulation Parameters

Radio Channel Parameters		**Traffic Model**	
Propagation model	Macro cell urban	UE Arrivals	Poisson Process with 2 users/min
Fading model	Pedestrian A	Connection Time	Exponential distribution with mean 60 seconds
Transmitted power	30 dBm	SLAs Λ	0.5 Mbps/UE, 1 Mbps/UE, 1.5 Mbps/UE
BS antenna gain	15 dBi	**SQR Parameters**	
Cell range	2 Km	P_{penalty}	2
Noise	9 dB	Randomness ϵ	0.1
Inference plus noise	-110 dBm	Learning Rate η	0.1
Carrier frequency	2 GHz	K nearest neighbors	10

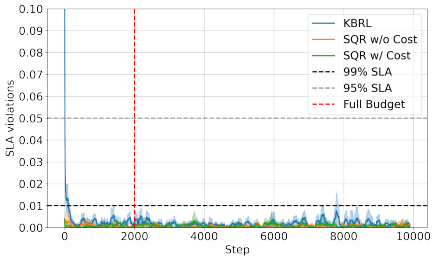


Fig. 2: SLA Violation Rate

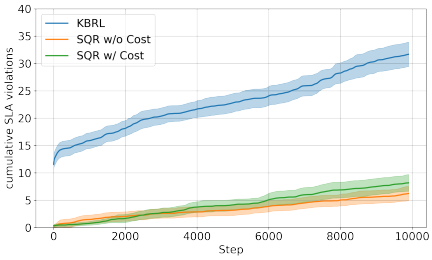


Fig. 3: Cumulative SLA Violation

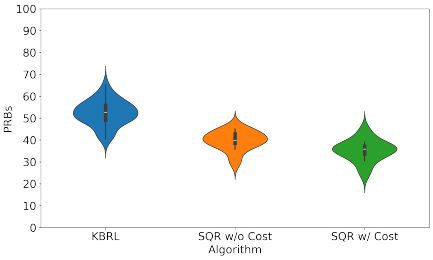


Fig. 4: Resource Allocation

safety. Thus, our method provides a principled mechanism for risk-averse adaptation without requiring explicit constraints or post-hoc correction.

V. Experimental Evaluation and Analysis

To evaluate the effectiveness of the proposed Safe Quantile Regression (SQR) agent, we conduct a comprehensive system-level simulation designed to emulate the dynamics of a realistic O-RAN deployment, in which the model is updated in every 100 slots and each update takes less than 100 ms.

Our simulation models a single macro-cell base station with a coverage radius of 2 km operating at a carrier frequency of 2 GHz. Radio propagation follows the 3GPP macro-cell urban path loss model [14], combined with the Pedestrian A fading profile [15] to capture realistic mobility-induced signal variation. The base station serves a mix of mobile users whose spatial positions and trajectories are randomly generated per episode. User arrivals follow a Poisson process, and each connection's duration is exponentially distributed to simulate varying session lengths. The system supports three eMBB slices, each defined by distinct SLA configurations in terms of minimum throughput thresholds. Each UE is randomly assigned to one of the slices and managed by the agent accordingly. We keep a support vector memory with 2000 entries and quantile rage as [0.01, 0.05]. Table I summarizes the full list of simulation and agent hyperparameters.

We evaluate the SQR agent under two configurations—with ($\gamma = 1$) and without ($\gamma = 0$) the cost-sensitive penalty mechanism—and compare both variants against KBRL [12], a model-based reinforcement learning baseline previously studied for O-RAN control. The evaluations are conducted over multiple runs to ensure statistical reliability, and all metrics are aggregated across the three slices to enable consistent cross-method comparison.

Figure 2 shows the average SLA violation rates over time. While all methods eventually converge to satisfy the 99% SLA target, key differences emerge during the early learning stages. Figure 3 presents the cumulative violation trends, where both SQR variants exhibit significantly flatter curves than KBRL, particularly during the initial phase. This demonstrates faster convergence and superior sample efficiency—critical advantages in online environments where early poor decisions can lead to prolonged user experience degradation. Notably, KBRL is evaluated under per-slice SLA constraints, whereas our method enforces SLA at the per-UE level, resulting in more stringent and granular guarantees. Consequently, KBRL tends to incur more violations due to its coarser enforcement granularity. Moreover, unlike approaches that rely on monolithic assumptions about network behavior, our model directly learns from fine-grained network outcomes, enabling more accurate SLA risk prediction and more reliable control.

A core challenge in online learning systems is bounded memory. In our setup, the support vector memory is capped at 2000 entries. Despite this constraint, the SQR agent continues to maintain high-quality predictions throughout training. This is achieved via an adaptive memory pruning strategy that discards low-impact or redundant samples, thereby maximizing the learning value of each stored support vector. The agent effectively learns which experiences are most valuable for future decision-making, demonstrating robustness under limited capacity.

Comparing the two SQR variants reveals a strategic trade-off introduced by the cost-sensitive penalty. The variant with penalties incurs slightly more SLA violations—typically

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within 0.003–0.008% range—but consumes fewer radio resources. This behavior is expected: the agent is incentivized to approach the SLA boundary more closely in pursuit of improved resource efficiency. The penalty mechanism drives the agent to make tighter, more aggressive decisions, occasionally increasing the risk of small SLA breaches but delivering significantly improved efficiency in return. Importantly, these violations are rare and minor, suggesting the penalty-based policy is well-calibrated for practical deployment.

Figure 4 presents the average number of RBs used over time. Both SQR agents achieve substantially lower RB consumption than KBRL, with the penalty-based SQR showing the highest efficiency—consistently saving over 60% in RB usage. This resource frugality is especially important in multi-slice environments where spectral resources are limited and must be shared across competing service demands. The results suggest that our approach not only meets service-level goals but also promotes globally efficient resource utilization.

Another important observation is the generalization capability of the SQR framework. Unlike KBRL, which relies on a monotonic relationship between actions and outcomes, our method does not impose such assumptions. This allows SQR to remain effective even under non-monotonic physical-layer effects, such as those caused by inter-cell interference, scheduling jitter, or abrupt mobility patterns. These real-world complexities often lead to counterintuitive behaviors where additional resources may not yield improved performance. The ability of our method to model such phenomena makes it better suited for deployment in production O-RAN systems.

VI. Conclusion and Future Work

This paper introduces a model-based reinforcement learning agent for O-RAN that achieves safe and efficient online learning. By leveraging online quantile regression, the agent explicitly models the SLA risk boundary and applies a cost-sensitive learning rule that promotes both safety and resource efficiency—without relying on strong assumptions about network dynamics. Experiments show that the proposed method reduces cumulative SLA violations by 60% and lowers radio resource consumption by 20% compared to baselines, striking a strong balance between safety and efficiency. These results highlight the potential of the SQR agent to enable intelligent, automated slice management in practical O-RAN scenarios.

As future work, we plan to validate our framework on real-world O-RAN testbeds to assess its robustness under practical conditions such as signaling latency, hardware constraints, and traffic variability. We aim to prototype the agent on large-scale 5G testbeds like Colosseum [16] and Arena [17] to evaluate its simulation-to-reality transfer, responsiveness, and safety. Furthermore, we will extend the agent to handle all three 5G slice types—eMBB, URLLC, and mMTC—to test its generalizability and develop adaptive inter-slice coordination mechanisms in multi-service O-RAN environments.

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